

CODEALPHA BIOINFORMATICS INTERNSHIP

TASK-3

Title:

3D Structure Prediction and Analysis of Human Lysozyme C

Protein Information

Protein name: **Haemoglobin subunit beta**

UniProt ID: **P68871**

Organism: Homo sapiens

Length: 147 amino acids

Function: Involved in oxygen transport from the lung to the various peripheral tissues

The screenshot displays the UniProt database entry for Hemoglobin subunit beta (P68871). The header includes the UniProt logo and navigation links like BLAST, Align, Peptide search, ID mapping, SPARQL, and UniProtKB. The main title is "P68871 · HBB_HUMAN". Below this, a table provides key information: Protein (Hemoglobin subunit beta), Gene (HBB), Status (UniProtKB reviewed (Swiss-Prot)), Organism (Homo sapiens (Human)), Amino acids (147), Protein existence (Evidence at protein level), and Annotation score (50). A sidebar on the left lists various categories: Function, Names & Taxonomy, Subcellular Location, Disease & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence, Similar Proteins, and Homologs. The main content area shows the "Function" section, which states: "Involved in oxygen transport from the lung to the various peripheral tissues." and "LVV-hemorphin-7 potentiates the activity of bradykinin, causing a decrease in blood pressure." It also mentions "Spinorphin" as an endogenous inhibitor of enkephalin-degrading enzymes. The "Miscellaneous" section notes that "One molecule of 2,3-bisphosphoglycerate can bind to two beta chains per hemoglobin tetramer." A "Caution" box is visible at the bottom.

Predicted Structure Features

The protein forms a compact globular structure.

Contains mainly **alpha-helices and beta-sheet regions**.

Has a conserved active-site region involved in catalytic activity.
Stabilized by disulfide bonds.
The structure contains a cleft where substrate binding occurs.

Structural Analysis

AlphaFold/SWISS-MODEL prediction shows a well-folded enzyme structure.

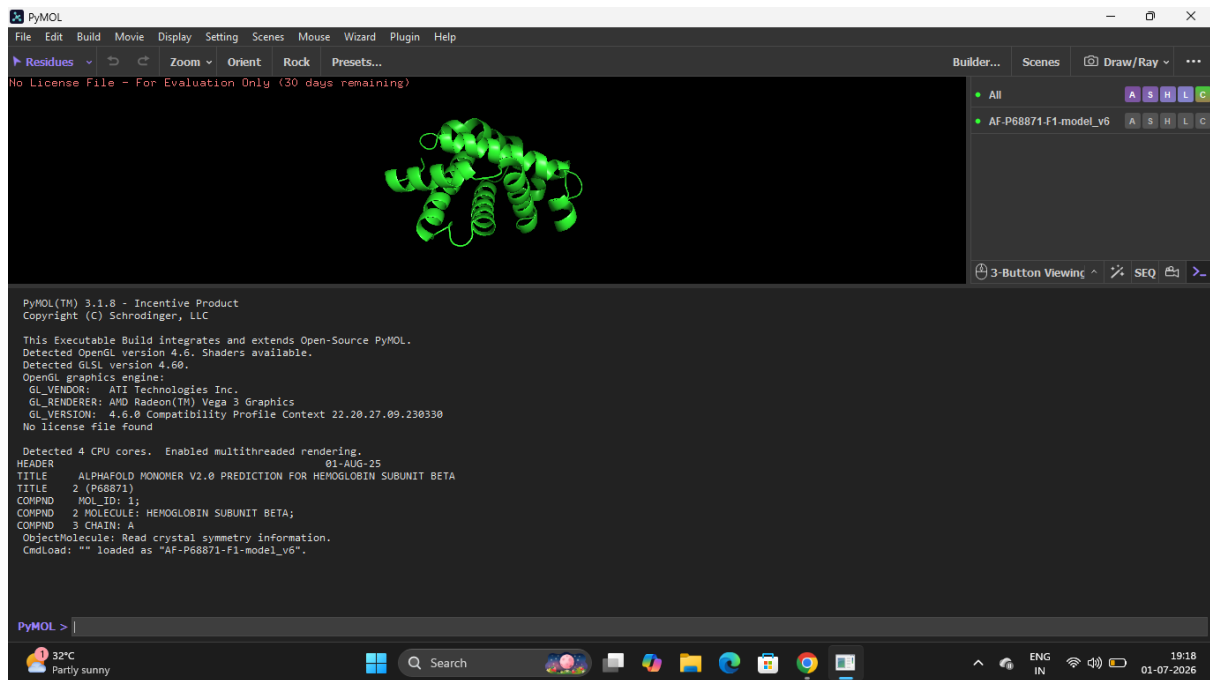
The screenshot shows the AlphaFold Protein Structure Database interface. At the top, there is a navigation bar with links: Home, About, FAQs, Downloads, API, and Collaborators. Below this is a search bar with the placeholder text "Search for protein, gene, UniProt accession or organism or sequence search". A "Search" button is located below the search bar. Below the search bar, there are examples of search terms: "MENFQKVEKIGEGTYGV...", "Free fatty acid receptor 2", "A11g58602", "Q911F6", and "E. coli". There are also links for "See search help" and "Go to online course".

The search results for "P68871" are displayed. It shows "Showing all search results for P68871" and "1 - 2 of 2 results". On the left, there is a "Filter by:" section with "Entry type" and "Status". Under "Entry type", there are links for "Monomers (1)", "Multimers (1)", and "Homodimers (1)". Under "Status", there is a link for "Review (?)".

On the right, there is a table of results. The first result is "Hemoglobin subunit beta". It is a "Monomer" from the "AF-P68871-F1-v6" dataset, which is part of the "Google DeepMind dataset". The table lists the following information:

Protein	Hemoglobin subunit beta
Gene	HBB
Source organism	<i>Homo sapiens</i> search this organism
UniProt	P68871 go to UniProt
Experimental structures	350 PDB structures for P68871 go to PDB-KB
Global quality	average pLDDT: 97.19.0 (very high)

The predicted model can be visualized using PyMOL.
Secondary structure elements contribute to protein stability and function.



Conclusion

The predicted 3D structure provides insight into the folding pattern and functional regions of **Haemoglobin subunit beta**. Structural visualization helps understand how the protein performs its biological role.